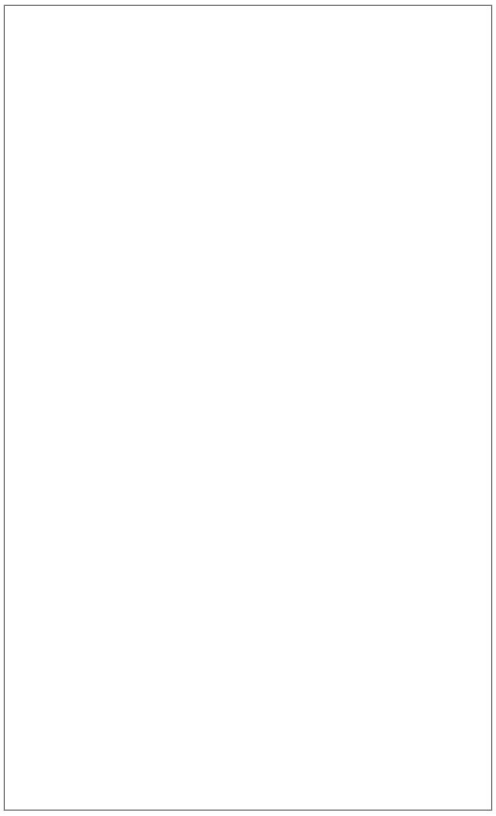
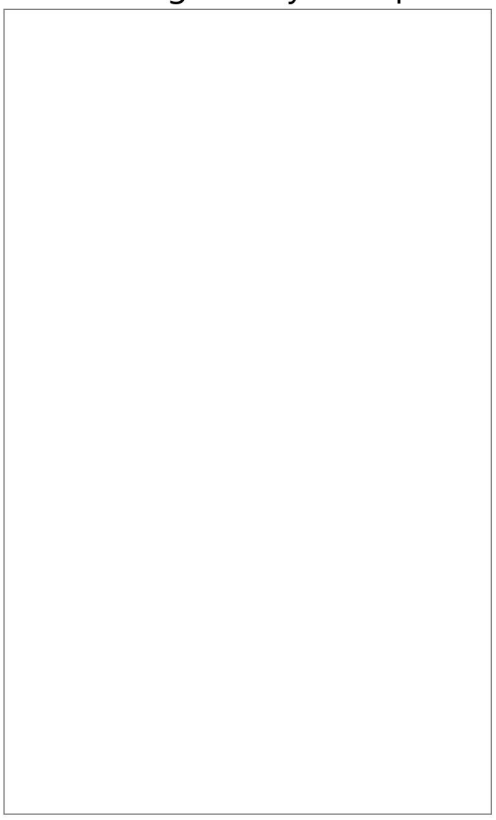
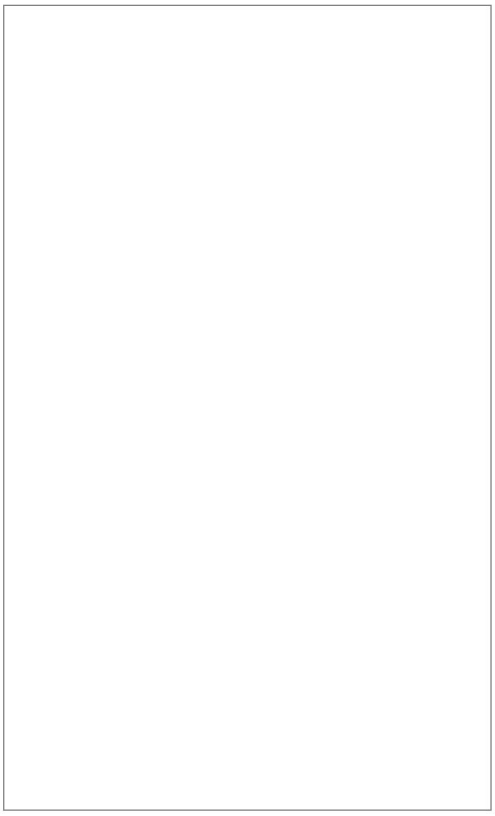
SAR Image Analysis Report



Original SAR (not found; preview shown)



Colorized



K-Means Segmented

Metadata:

- Original File: Not available
- Colorized File: colorized_1763463709.png
- Processed On: 2025-11-18 16:32:03 IST
- SAR Classifier Confidence: 0.000
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Vegetation (Green Tone)
Mean RGB: (169, 173, 170)
Area: 14.63% of image

Cluster 1: Likely Vegetation (Green Tone)
Mean RGB: (70, 88, 83)
Area: 32.95% of image

Cluster 2: Mixed / Other Surface
Mean RGB: (23, 31, 34)
Area: 19.95% of image

Cluster 3: Likely Vegetation (Green Tone)
Mean RGB: (115, 128, 123)
Area: 32.47% of image

Land Cover Summary:

- Likely Vegetation (Green Tone) → 14.63%
- Likely Vegetation (Green Tone) → 32.95%
- Mixed / Other Surface → 19.95%
- Likely Vegetation (Green Tone) → 32.47%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely vegetation (green tone) (80.05% combined across clusters).

Secondary presence: mixed / other surface (19.95%).

K-Means clustering groups pixels by color similarity — labels are heuristic approximations.

No direct spectral indices (e.g., NDVI) were computed; consider adding multispectral data for precise vegetation analysis.

Recommendations:

The analysis shows dominant land cover is vegetation (33.33%). Strong vegetation signal suggests this area is suitable for agricultural monitoring — consider crop health and biomass monitoring. Secondary clusters show distributed variations; using time-series imagery would help confirm seasonal changes. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.000
Dominant Class Percentage: 32.95%
Classification Reliability: High

Limitations:

- KMeans clustering is unsupervised and can group visually similar pixels that are not the same land-cover type.
- GAN-based colorization is a synthetic approximation and does not replace real multispectral imagery.
- SAR classifier confidence depends on training data and input SAR quality; treat results as indicative, not
- definitive.
- Reports rely on single-image snapshot; multi-temporal analysis improves robustness.

Final Summary:

The generated analysis indicates the region is predominantly vegetation-dominated with some mixed surfaces.

No significant water bodies or dense urban zones were detected in this image.

Recommend periodic monitoring and incorporation of ground-truth data for higher confidence.